

Experiment 10:

Aim: To demonstrate the performance of Logistic Regression using iris dataset

Program:

```
# Import libraries
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Load Iris dataset
iris = load_iris()

# Features and Target
X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=200)

# Train model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:\n")
print(cm)

# Classification Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# Plot Confusion Matrix
plt.figure(figsize=(6,5))
sns.heatmap(cm,
            annot=True,
            fmt='d',
            cmap='Blues',
```

```
xticklabels=iris.target_names,  
yticklabels=iris.target_names)
```

```
plt.title("Logistic Regression - Iris Dataset")  
plt.xlabel("Predicted")  
plt.ylabel("Actual")  
plt.show()
```

Accuracy: 1.0

Confusion Matrix:

```
[[19  0  0]  
 [ 0 13  0]  
 [ 0  0 13]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	19
1	1.00	1.00	1.00	13
2	1.00	1.00	1.00	13
accuracy			1.00	45
macro avg	1.00	1.00	1.00	45
weighted avg	1.00	1.00	1.00	45

